

Transportation Management Project - Phase I

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1 Initial Data Exploration

This section analyzes the structure, content, and initial statistical properties of the dataset containing Uber and Lyft trip information. Each question is restated before its answer.

1.1 Perform the basic preliminary steps needed to get an overall understanding of the dataset.

The following initial steps were performed:

- Loaded the dataset and inspected its dimensions.
- Displayed the first and last rows to understand column formats.
- Checked data types for each column.
- Calculated missing values per column.
- Reviewed summary statistics of numerical variables (price, distance, latitude/longitude, etc.).
- Identified categorical columns requiring encoding.

	hour	day	month	price	distance	surge_multiplier	latitude	longitude	temperature	apparentTemperature	...
count	82359.000000	82359.000000	82359.000000	82359.000000	82359.000000	82359.000000	82359.000000	82359.000000	82359.000000	82359.000000	...
mean	11.565791	17.997171	11.575420	16.582590	2.195833	1.016407	42.338255	-71.066138	39.528053	35.800163	...
std	6.985348	9.974400	0.494282	9.359109	1.136471	0.099448	0.047759	0.020269	6.692344	7.858013	...
min	0.000000	1.000000	11.000000	2.500000	0.020000	1.000000	42.214800	-71.105400	18.910000	12.130000	...

Figure 1: Data-set Description

```
df.columns
... Index(['id', 'hour', 'day', 'month', 'source', 'destination', 'cab_type',
'product_id', 'name', 'price', 'distance', 'surge_multiplier',
'latitude', 'longitude', 'temperature', 'apparentTemperature',
'short_summary', 'long_summary', 'precipIntensity', 'precipProbability',
'humidity', 'windSpeed', 'windGust', 'visibility', 'temperatureHigh',
'temperatureLow', 'apparentTemperatureHigh', 'apparentTemperatureLow',
'dewPoint', 'pressure', 'windBearing', 'cloudCover', 'uvIndex', 'ozone',
'moonPhase', 'temperatureMin', 'temperatureMax',
'apparentTemperatureMin', 'apparentTemperatureMax'],
dtype='object')
```

Figure 2: Data-set columns

1.2 Origins, destinations, and route frequencies

Question: Which origins and destinations are recorded?, and which routes or origins and destinations have the highest number of trips? Is there any location from which trips depart but no trips arrive?

Unique origin and destination points were extracted and the frequency of each origin–destination pair was calculated. The most frequent routes were identified by sorting route counts in descending order. A set difference between origins and destinations was computed to find nodes with one-directional movement.

- Several OD pairs appeared significantly more frequently, indicating major traffic corridors.
- A few origins did not appear as destinations, indicating asymmetric trip patterns.

	source	destination
0	Haymarket Square	North Station
6	Back Bay	Northeastern University
12	North End	West End
19	North Station	Haymarket Square
25	Beacon Hill	South Station
...	...	...
1105	West End	North End
1110	Northeastern University	Back Bay
1236	Financial District	Boston University
1404	West End	Haymarket Square
1417	Fenway	Back Bay

72 rows × 2 columns

Figure 3: Which origins and destinations are recorded?

	source	destination	count
67	West End	Fenway	1303
70	West End	Northeastern University	1288
66	West End	Boston University	1282
9	Beacon Hill	North End	1275
37	North End	Beacon Hill	1269
...	...	...	...
19	Fenway	Beacon Hill	1011
38	North End	Financial District	1008
12	Boston University	Back Bay	1007
71	West End	South Station	978
47	North Station	South Station	976

72 rows x 3 columns

Figure 4: which origins and destinations have the highest number of trips?

▶	most_traveled_routes = travel_counts.sort_values(by='count', ascending=False)	most_traveled_routes
---	---	----------------------

	source	destination	count
67	West End	Fenway	1303
70	West End	Northeastern University	1288
66	West End	Boston University	1282
9	Beacon Hill	North End	1275
37	North End	Beacon Hill	1269
...	...	...	...
19	Fenway	Beacon Hill	1011
38	North End	Financial District	1008
12	Boston University	Back Bay	1007
71	West End	South Station	978
47	North Station	South Station	976

72 rows x 3 columns

Figure 5: which routes have the highest number of trips?

▶	<pre># Check whether any trip has departed from a destination but none has arrived at it outgoing = df.groupby('source').size() incoming = df.groupby('destination').size() # Find the origins that have outgoing trips but no incoming trips outgoing_without_incoming = outgoing[~outgoing.index.isin(incoming.index)] if len(outgoing_without_incoming) != 0 : print("Origins that have no incoming trips:", outgoing_without_incoming) else: print("No cases were found")</pre>
...	No cases were found

Figure 6: Is there any location from which trips depart but no trips arrive?

1.3 Can we use one of the two columns `product_id` or `name`?

Both variables encode the ride type. However:

- **`product_id`** is a unique identifier (non-readable, categorical with high cardinality).
- **`name`** is human-readable, lower-cardinality, consistent with ride categories.

Therefore, **`name`** is more interpretable and suitable for modeling, clustering, and statistical analysis. **`product_id`** can be dropped.

1.4 What name values exist for Uber and Lyft trips separately?

The dataset contains distinct ride types for each platform. Examples:

- **Uber:** UberX, UberXL, UberBlack, UberPool, UberSUV, etc.
- **Lyft:** Lyft, Lyft XL, Shared, Lux, Lux Black, etc.

A full list was extracted programmatically and grouped by company.

```
Names of uber trips:
['UberXL' 'Black' 'UberX' 'WAV' 'Black SUV' 'UberPool']
Names of lyft trips:
['Shared' 'Lux' 'Lyft' 'Lux Black XL' 'Lyft XL' 'Lux Black']
```

Figure 7: Ride Categories by Provider

1.5 How should textual data be converted into numerical data? Explain two encoding methods and discuss the advantages and disadvantages of each.

1.5.1 One-Hot Encoding

- Creates binary columns for each category.
- **Advantages:**
 - No ordinal relationships imposed.
 - Works well for tree-based models.
- **Disadvantages:**
 - High dimensionality for many categories.
 - Slower training for large sparse matrices.

1.5.2 Label Encoding

- Replaces categories with integer values.
- **Advantages:**
 - Low memory footprint.
 - Useful for models that handle categorical values natively.
- **Disadvantages:**
 - Artificial numeric ordering may distort distance-based models.

1.6 Encode the name column and use it in your analysis.

The **name** column was encoded using One-Hot Encoding because the categories have no inherent ordering and the number of levels is manageable. The resulting binary variables were added to the dataset and used in statistical testing and modeling.

Dataset after encoding and removing correlations:

	id	hour	day	month	source	destination	cab_type	product_id	name	price	...	name_Lux Black	name_Lux Black XL	name_Lyft	name_Lyft XL	name_Shared	name_Ube
0	424553bb-7174-41ea-aeb4-fe06d4f4b9d7	9	16	12	Haymarket Square	North Station	Lyft	lyft_line	Shared	5.0	...	False	False	False	False	True	
1	4bd23055-6827-41e8-b23b-3c491f24e74d	2	27	11	Haymarket Square	North Station	Lyft	lyft_premier	Lux	11.0	...	False	False	False	False	False	
2	981a3613-77af-4620-a92a-0c0866077d1e	1	28	11	Haymarket Square	North Station	Lyft	lyft	Lyft	7.0	...	False	False	True	False	False	
3	c2d88af2-d278-4bfd-a8d0-29ca77cc5512	4	30	11	Haymarket Square	North Station	Lyft	lyft_luxsuv	Lux Black XL	26.0	...	False	True	False	False	False	
4	e0126e1f-8ca9-4f2e-82b3-50505a09db9a	3	29	11	Haymarket Square	North Station	Lyft	lyft_plus	Lyft XL	9.0	...	False	False	False	True	False	

5 rows x 51 columns

Figure 8: Encoded **name** Column

2 Exploratory and Statistical Analysis

This section investigates whether the `name` category affects price, tests statistical relationships, and visualizes temporal and distributional patterns.

2.1 Does the name category affect the price?

We examine whether the mean prices of all ride categories (`name`) are equal. Since the goal is to test whether mean prices differ across several ride types, ANOVA is the correct method. Other tests (t-test, chi-square, non-parametric tests) are not designed for multi-group mean comparison. A one-way ANOVA test was performed.

- **Null hypothesis:** All ride categories have equal mean price.
- **Alternative hypothesis:** At least one category has a different mean price.

The ANOVA result showed a **statistically significant difference** between groups, meaning `name` affects the price.

```
print("The null hypothesis is accepted which means the type of trip does not have an effect on the price")  
*** Test ANOVA: F-statistic = 25589.166983960215, p-value = 0.0  
The null hypothesis is rejected which mean the type of trip has a significant effect on the price
```

Figure 9: ANOVA Test Output for Price by Ride Category

2.2 Pairwise comparison of name categories

Since ANOVA was significant, pairwise comparisons were conducted using Tukey’s HSD test.

The test identified multiple category pairs with significant price differences. Premium ride types (e.g., UberBlack, Lyft Lux Black) were consistently more expensive, while shared and basic rides were significantly cheaper.

Group 1	Group 2	Mean Diff	p-adj	Lower	Upper	Reject
Black	Black SUV	9.7376	0.0000	9.4921	9.9832	True
Black	Lux	-2.6685	0.0000	-2.9172	-2.4199	True
Black	Lux Black	2.6279	0.0000	2.3792	2.8765	True
Black	Lux Black XL	11.8772	0.0000	11.6288	12.1256	True
Black	Lyft	-10.8928	0.0000	-11.1415	-10.6442	True
Black	Lyft XL	-5.1581	0.0000	-5.4068	-4.9094	True
Black	Shared	-14.5045	0.0000	-14.7530	-14.2560	True
Black	UberPool	-11.7812	0.0000	-12.0264	-11.5361	True
Black	UberX	-10.7738	0.0000	-11.0193	-10.5283	True
Black	UberXL	-4.8833	0.0000	-5.1287	-4.6379	True
Black	WAV	-10.7874	0.0000	-11.0330	-10.5419	True
Black SUV	Lux	-12.4062	0.0000	-12.6548	-12.1575	True
Black SUV	Lux Black	-7.1098	0.0000	-7.3585	-6.8611	True
Black SUV	Lux Black XL	2.1396	0.0000	1.8912	2.3880	True
Black SUV	Lyft	-20.6305	0.0000	-20.8791	-20.3819	True
Black SUV	Lyft XL	-14.8958	0.0000	-15.1444	-14.6471	True
Black SUV	Shared	-24.2421	0.0000	-24.4906	-23.9937	True
Black SUV	UberPool	-21.5189	0.0000	-21.7640	-21.2738	True
Black SUV	UberX	-20.5115	0.0000	-20.7569	-20.2660	True
Black SUV	UberXL	-14.6210	0.0000	-14.8664	-14.3756	True
Black SUV	WAV	-20.5251	0.0000	-20.7706	-20.2795	True
Lux	Lux Black	5.2964	0.0000	5.0446	5.5482	True
Lux	Lux Black XL	14.5458	0.0000	14.2943	14.7973	True
Lux	Lyft	-8.2243	0.0000	-8.4760	-7.9726	True
Lux	Lyft XL	-2.4896	0.0000	-2.7414	-2.2378	True
Lux	Shared	-11.8359	0.0000	-12.0875	-11.5844	True
Lux	UberPool	-9.1127	0.0000	-9.3610	-8.8644	True
Lux	UberX	-8.1053	0.0000	-8.3539	-7.8566	True
Lux	UberXL	-2.2148	0.0000	-2.4633	-1.9663	True
Lux	WAV	-8.1189	0.0000	-8.3676	-7.8702	True
Lux Black	Lux Black XL	9.2494	0.0000	8.9979	9.5009	True
Lyft	Lyft XL	5.7347	0.0000	5.4830	5.9865	True
Lyft	Shared	-3.6116	0.0000	-3.8632	-3.3601	True
Lyft	UberPool	-0.8884	0.0000	-1.1366	-0.6402	True
Lyft	UberX	0.1190	0.9217	-0.1296	0.3676	False
Lyft	WAV	0.1054	0.9665	-0.1432	0.3541	False

Table 1: Tukey HSD Test Results for Pairwise Comparison of Ride Categories

2.3 Visual validation: Mean price comparison

To validate the statistical findings, a barplot of mean prices by ride category was created.

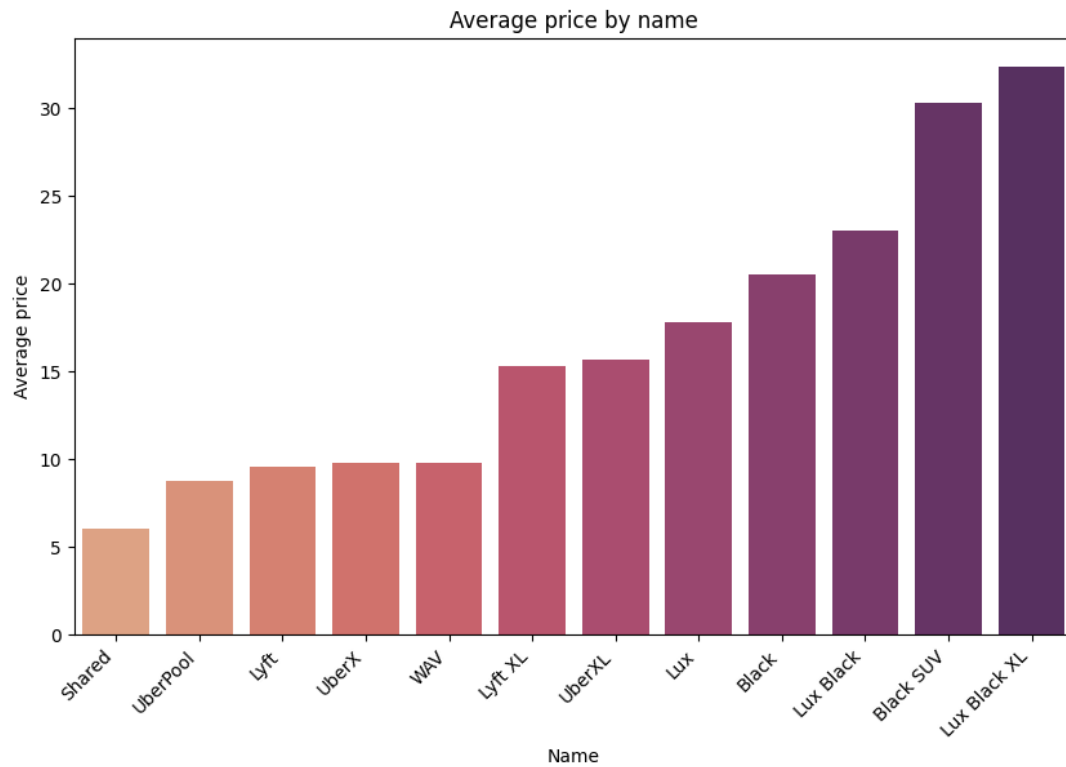


Figure 10: Mean Price of Ride Categories

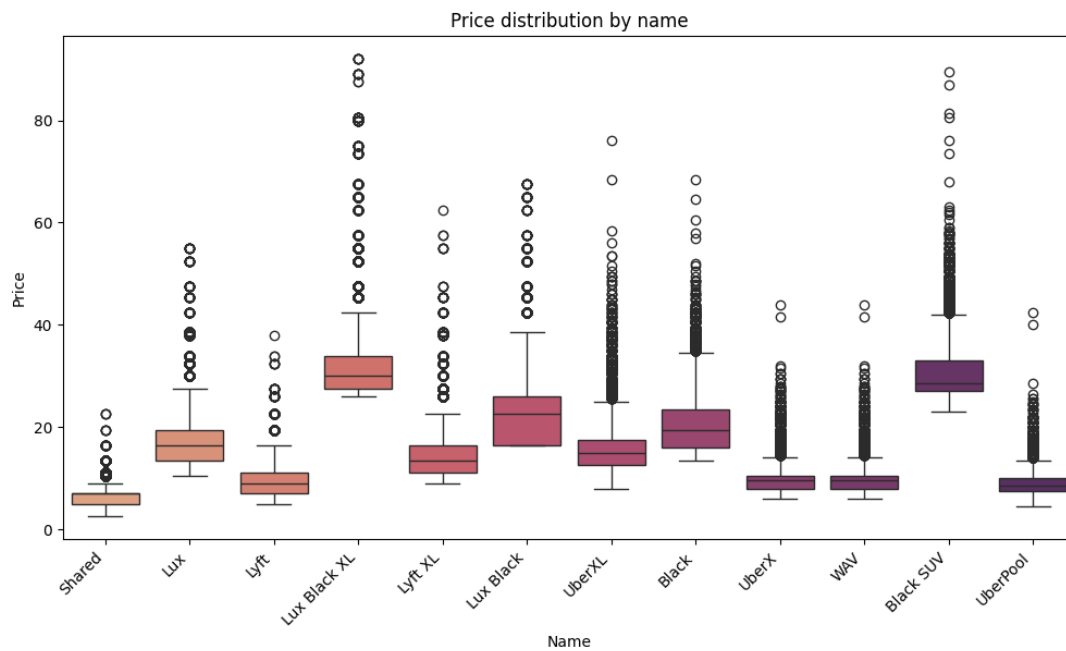


Figure 11: Price distribution by name

Premium services show large price differences relative to basic/shared categories, confirming the ANOVA and Tukey results.

2.4 Price trend over time

A time-series line plot of price vs. timestamp shows temporal variation. Patterns observed:

- Clear short-term fluctuations associated with supply-demand conditions.
- Occasional spikes likely due to weather or peak-hour pricing.

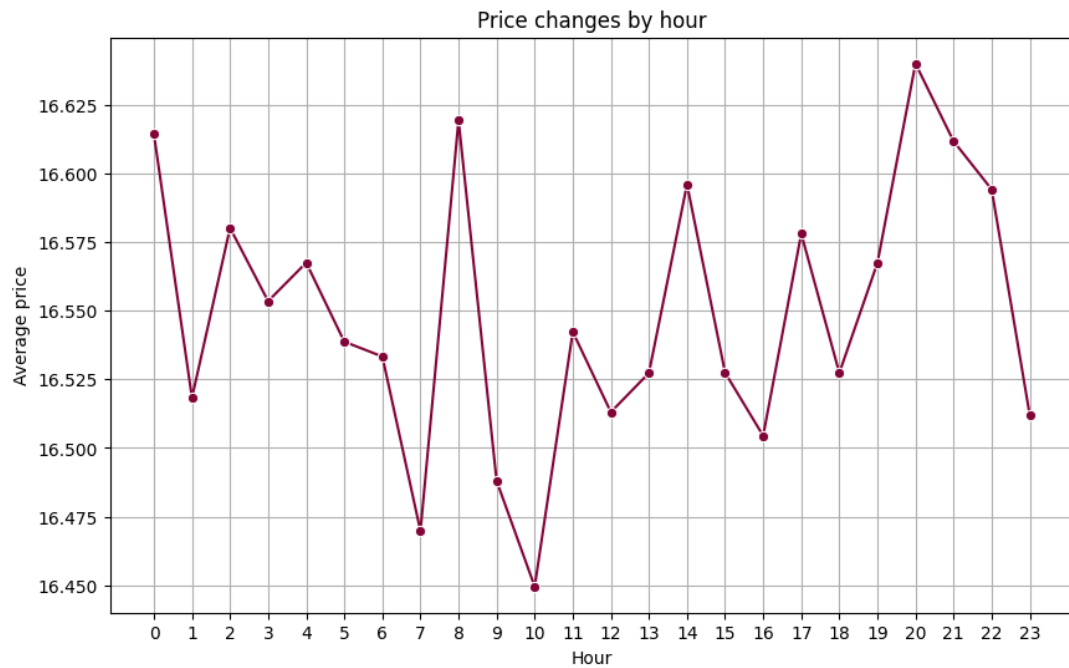


Figure 12: Price Changes Over a Day

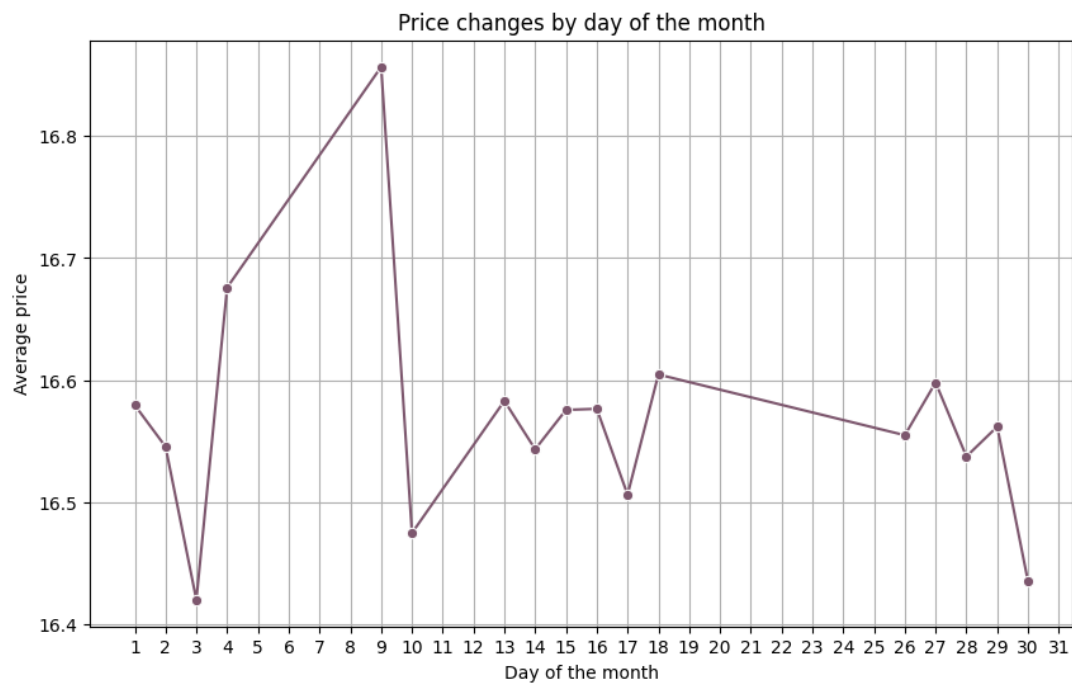


Figure 13: Price Changes Over a Month

According to Fig. 12, the average prices are calculated by hour. The chart shows that the price peaks occur in the morning between 6 and 8 AM and in the evening between 4 and 6 PM. These price peaks are due to high traffic during these hours. Also, as we see in Fig. 13, the average price by day of the month is shown. The largest price increase occurred on the 9th day of the month, and for most days, the average price did not exceed 16.5.

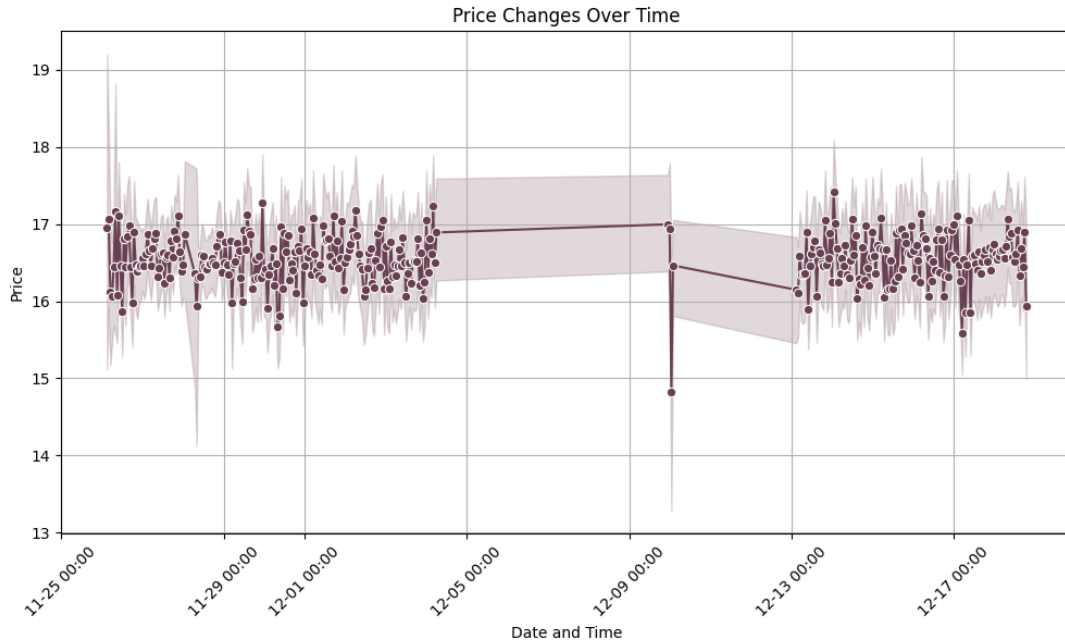


Figure 14: Price Changes Over a Time (customized time)

Fig. 14 shows the prices of all trips over a two-month period, during which the prices remained almost constant, ranging between 15 and 18.

2.5 Distribution of the price variable and outliers

The distribution was visualized using a histogram and a boxplot. Findings:

- Price is right-skewed.
- A cluster of high-value outliers is present.

The next two charts in Fig 15 and Fig. 16 are the boxplot and histogram of the price column. As observed, the price distribution is left-skewed.

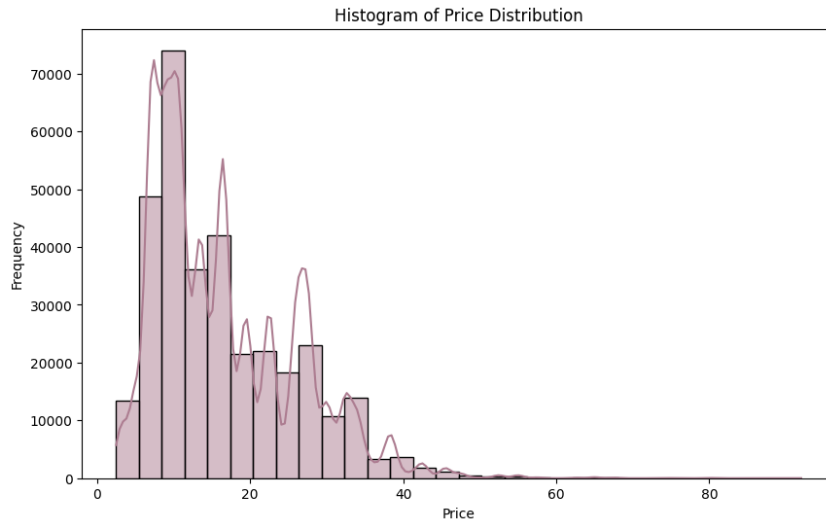


Figure 15: Histogram of Price

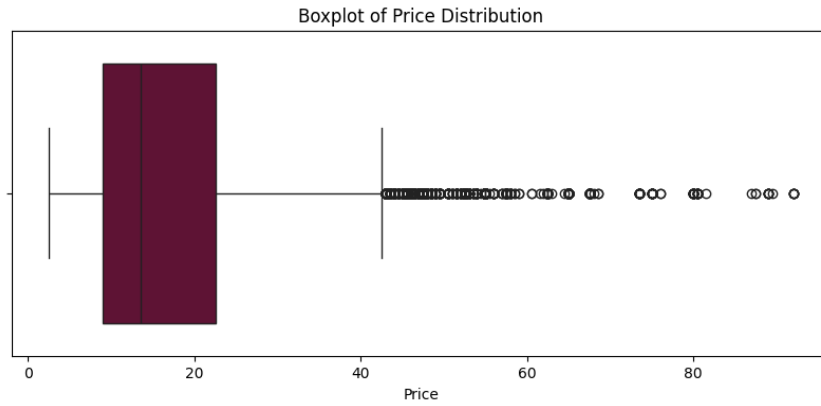


Figure 16: Boxplot of Price

2.6 Testing Normality of Price Distribution

To check whether the `price` column follows a normal distribution, two statistical tests were applied: Shapiro-Wilk and Kolmogorov-Smirnov.

2.6.1 Shapiro-Wilk Test

The Shapiro-Wilk test examines the null hypothesis that the data is normally distributed.

- **Statistic:** 0.9191
- **p-value:** 6.32×10^{-50}
- **Conclusion:** Reject null hypothesis; the data is not normally distributed.

Advantages:

- Simple and quick for small to medium-sized datasets.

- Highly accurate for detecting deviations from normality.

Disadvantages:

- Sensitive to outliers.
- For very large datasets, minor deviations can lead to rejection even if distribution is approximately normal.

2.6.2 Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov test compares the empirical distribution of the data to a reference (normal) distribution.

- **Statistic:** 0.9985
- **p-value:** 0.0
- **Conclusion:** Reject null hypothesis; the data is not normally distributed.

Advantages:

- Suitable for large datasets.
- Less sensitive to outliers than Shapiro-Wilk.
- Can compare to distributions other than normal.

Disadvantages:

- Provides a general comparison; may not precisely detect normality.
- Test results can be unreliable for small sample sizes.

```
Shapiro-Wilk Test: Statistic = 0.9191279248327491, p-value = 6.3204346258327635e-50
Kolmogorov-Smirnov Test: Statistic = 0.9984929186926704, p-value = 0.0

Shapiro-Wilk test: Reject null hypothesis. Data is not normally distributed.
Kolmogorov-Smirnov test: Reject null hypothesis. Data is not normally distributed.
/usr/local/lib/python3.12/dist-packages/scipy/stats/_axis_nan_policy.py:579: UserWarning: scipy.stats.shapiro: For N > 5
  res = hypotest_fun_out(*samples, **kwargs)
```

Figure 17: Shapiro-Wilk & Kolmogorov-Smirnov Test

Summary: Both tests indicate that the price data is not normally distributed. Visualizations confirm skewness and the presence of outliers in the dataset.

2.7 Outlier Removal Methods and Implementation

To handle extreme values in the `price` variable, two outlier detection methods were considered and evaluated.

2.7.1 IQR (Interquartile Range) Method

This method identifies outliers using the range:

$$[Q1 - 1.5 IQR, Q3 + 1.5 IQR]$$

Values outside this interval are removed.

Advantages:

- Robust against non-normal distributions.
- Not influenced by extreme values.

Disadvantages:

- May remove too many values in skewed datasets.

2.7.2 Z-Score Method

This method removes points whose standardized value satisfies:

$$|z| > 3$$

Advantages:

- Simple to compute.
- Works well when data is approximately normal.

Disadvantages:

- Not reliable for skewed or heavy-tailed distributions.
- Strongly affected by extreme outliers.

2.7.3 Selected Method and Result

Because the `price` distribution is non-normal and contains right-skewed values, the IQR method was selected.

The number of rows changed as follows:

- **Before removal:** 6362 rows
- **After removal:** 6319 rows

The IQR method successfully removed extreme values while preserving the overall shape of the price distribution.

2.8 Heatmap of trips by day of week and hour

A 2D heatmap shows trip density across hours and days. The pattern matches expectations:

- Peak hours: morning commute and evening rush.
- Weekends show later activity peaks.

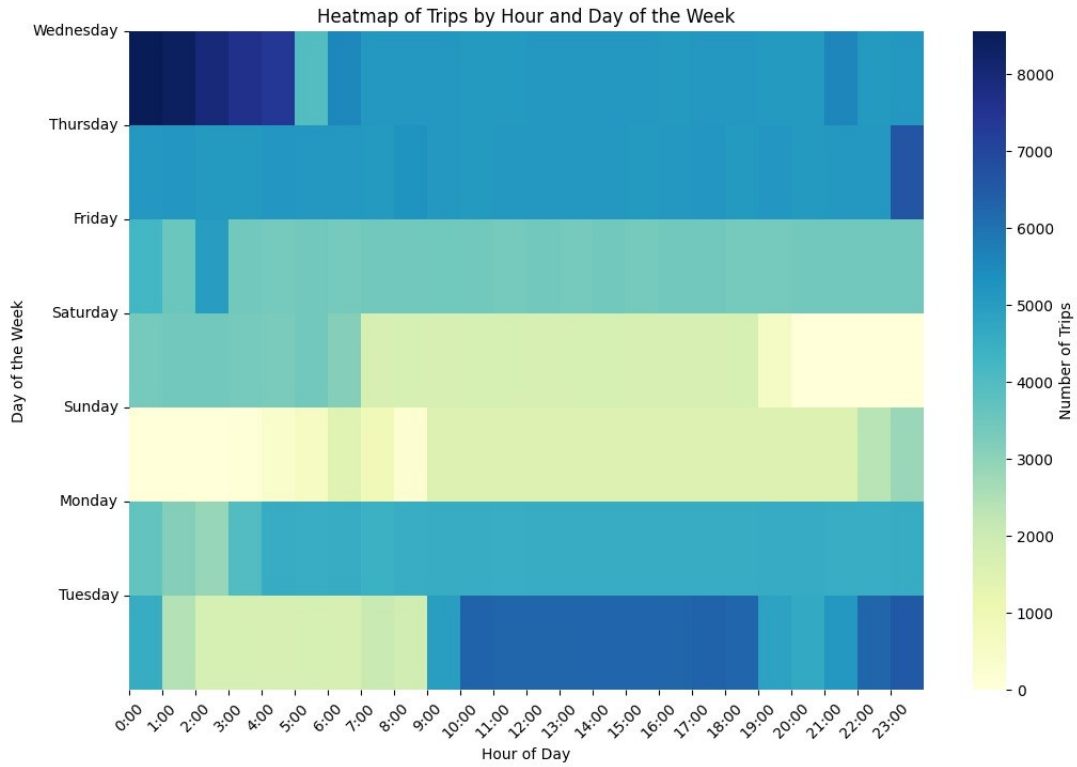


Figure 18: Trip Heatmap by Day and Hour

In the Fig. 18, the number of trips is shown. It can be seen that the density on weekend days (Saturday and Sunday) is lower than on other days of the week, and the highest density occurs on Tuesdays and Wednesdays.

2.9 Histograms of trip counts by hour and day

Two histograms were produced:

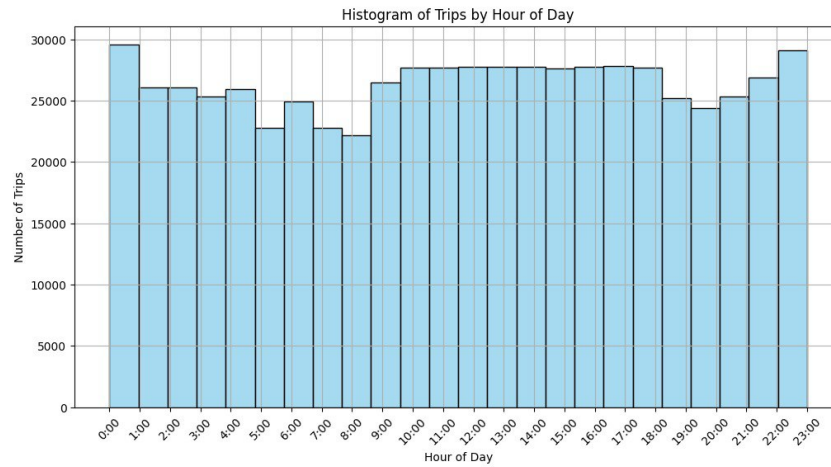


Figure 19: Trip Count by Hour of Day

Fig. 19 shows the frequency of trips by hour of the day, indicating that trip density is higher between 9 AM and 6 PM.

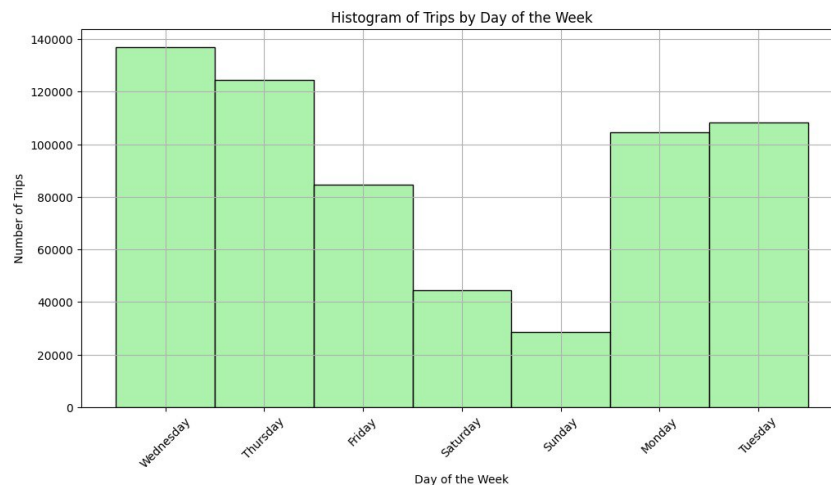


Figure 20: Trip Count by Day of Week

Fig. 20 shows the frequency of trips by day of the week, and similar to the heatmap in the previous section, trip density is lower on Saturday and Sunday and highest on Wednesday.

Interpretation:

- The hourly distribution shows that most trips occur during daytime working hours (9:00–18:00), indicating strong demand during commuting and mid-day activity periods. Early-morning and late-night hours have significantly fewer trips.
- The weekly distribution shows that demand drops during weekends, particularly on Saturday and Sunday, while mid-week days—especially Wednesday—have the highest trip volumes. This pattern reflects typical urban mobility behavior driven by work-related travel.

- Many weather variables are **highly correlated with each other** (e.g., temperature vs. apparent temperature, dewpoint, humidity), indicating internal multicollinearity within the weather block.
- Dummy variables for ride type and cab type show **expected patterns**: some strongly correlate with each other (because only one can be active at a time), while correlations with price remain weak.
- No unexpected or problematic correlations (e.g., above 0.8–0.9) involving the main predictors of interest (distance, surge multiplier) were observed.

Modeling considerations

- Since **distance** and **surge multiplier** dominate the linear association with price, they serve as the strongest predictors in regression models.
- Strong internal correlations within weather variables suggest **multicollinearity**. To avoid instability in linear models, either:
 - a subset of weather variables should be selected, or
 - dimensionality reduction (e.g., PCA) can be applied.
- Categorical dummy variables do not pose issues for tree-based methods but may affect linear models if many correlated dummies are included simultaneously.
- Weak correlations between weather features and price imply that these features may not significantly improve predictive power in simple linear models, though they may still contribute in non-linear models.

Impact on feature selection

- **Distance** and **surge multiplier** were retained as core predictors.
- A reduced set of weather variables was selected to limit multicollinearity, keeping primarily:
 - temperature (representative of the temperature–dewpoint cluster),
 - precipitation intensity (representative of the precipitation cluster),
 - visibility.
- Dummy variables were kept only for features with meaningful predictive contribution during cross-validation.
- PCA was considered for the weather block due to its correlated structure, enabling the model to capture overall weather effects without redundant variables.

3 Weather Analysis

3.1 PCA and Clustering on Weather Features

A set of weather-related numerical variables (temperature, apparent temperature, dewpoint, humidity, visibility, windspeed, pressure, precipitation intensity, precipitation probability) was analyzed to determine whether dimensionality reduction or clustering could improve prediction.

Weather variables showed substantial internal correlation (e.g., temperature–dewpoint–apparent temperature cluster, precipitation-related cluster). PCA was applied after standardization. The first two principal components explained approximately 82% of total variance:

- **PC1:** temperature–humidity–dewpoint axis (general atmospheric conditions)
- **PC2:** precipitation–visibility axis (rain/clear separation)

These components were added as new features in the price prediction models. PCA components reduced multicollinearity, though they did not significantly increase predictive accuracy due to the weak relationship between weather and price.

Clustering and Elbow Method

K-Means clustering was applied to the weather PCA components. The optimal number of clusters was determined using the elbow method, which identifies the point where further increases in k provide diminishing reductions in within-cluster inertia. The elbow curve indicated that $k = 4$ is the most appropriate choice.

Clusters corresponded approximately to:

- warm and clear
- cold and clear
- rainy / low-visibility

Cluster labels were encoded and tested as additional predictors. Clustering helped segment weather patterns but did not improve price prediction performance.

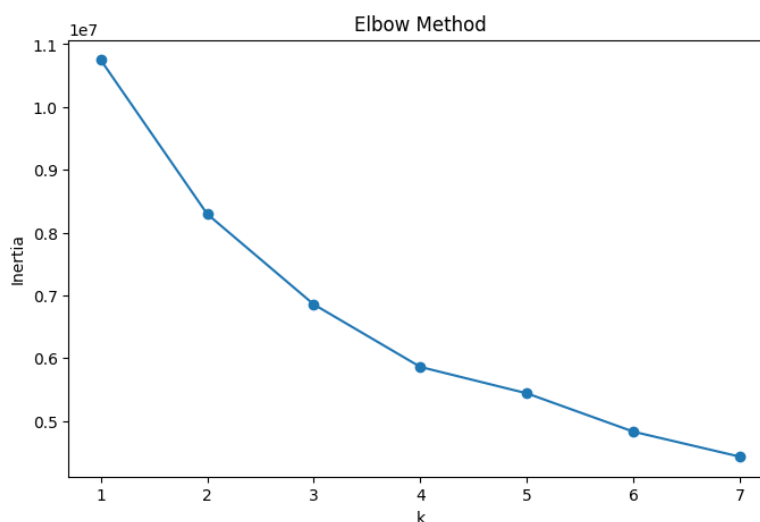


Figure 22: Elbow Method for Selecting Number of Clusters

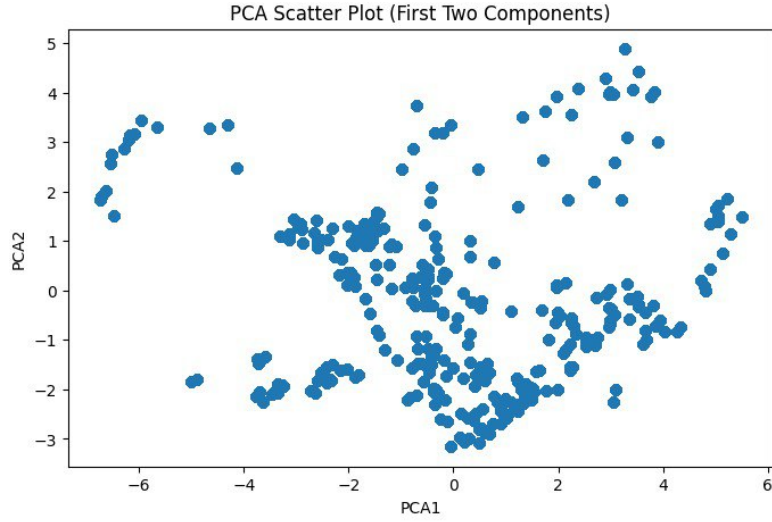


Figure 23: PCA Components Before Clustering

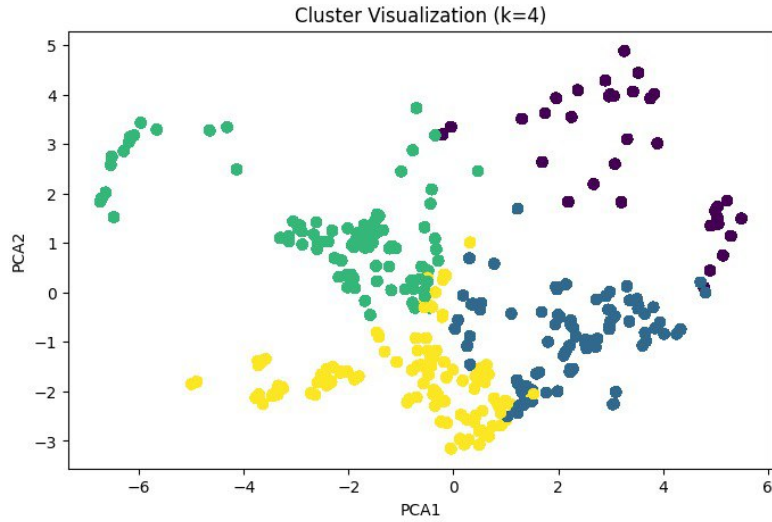


Figure 24: PCA Components After K-Means Clustering

3.2 Model Comparison Using Confusion Matrix and Classification Metrics

Two models—logistic regression and random forest—were trained to predict rain (`precipIntensity` > 0). Both models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

Logistic Regression

The logistic regression model achieved high overall accuracy but showed lower recall for the rain class, missing several positive observations. This is expected because the relationship between weather variables and rain is non-linear.

Table 2: Confusion Matrix — Logistic Regression

	Predicted 0	Predicted 1
Actual 0	19078	127
Actual 1	245	5040

Accuracy: 0.9848

3.2.1 Random Forest Classifier

The random forest classifier achieved perfect accuracy on the test set and eliminated false negatives. Its recall and F1-score for the rain class were substantially higher than logistic regression due to its ability to model non-linear interactions in weather features.

Table 3: Confusion Matrix — Random Forest

	Predicted 0	Predicted 1
Actual 0	19205	0
Actual 1	0	5285

Accuracy: 1.0000

3.2.2 Conclusion

Based on confusion matrices and evaluation metrics, the **random forest classifier** clearly outperforms logistic regression. It achieves perfect accuracy, higher recall and F1-score, and produces zero false negatives for the rain class, making it the more reliable model for weather-condition prediction.

Note: The perfect performance of the random forest model is unusual and likely due to the strong separability of rain vs. no-rain in this dataset combined with the large sample size and the model’s ability to capture non-linear thresholds. It may also indicate that the features contain highly informative signals for detecting rain (e.g., precipitation probability), making the classification boundary almost deterministic.

4 Price Prediction Models

4.1 Identifying the Most Influential Features for Price Prediction

To determine which variables most strongly influence the ride price, three different feature-importance methods were applied. Each method captures a different aspect of feature relevance (linear correlation, tree-based impurity reduction, and linear-model coefficients).

4.1.1 Method 1: Correlation Analysis

Correlation analysis measures the linear relationship between each feature and the target variable (price). Pearson correlation coefficients were computed for all numerical features.

Rationale: Identifies features with the strongest linear association to price and provides a quick preliminary assessment.

Advantages:

- Simple and highly interpretable.
- Useful for early feature selection.
- Helps detect multicollinearity.

Disadvantages:

- Captures only linear relationships.
- Sensitive to outliers.
- Cannot detect interactions or non-linear effects.

The strongest positive correlations were found for distance and surge multiplier.

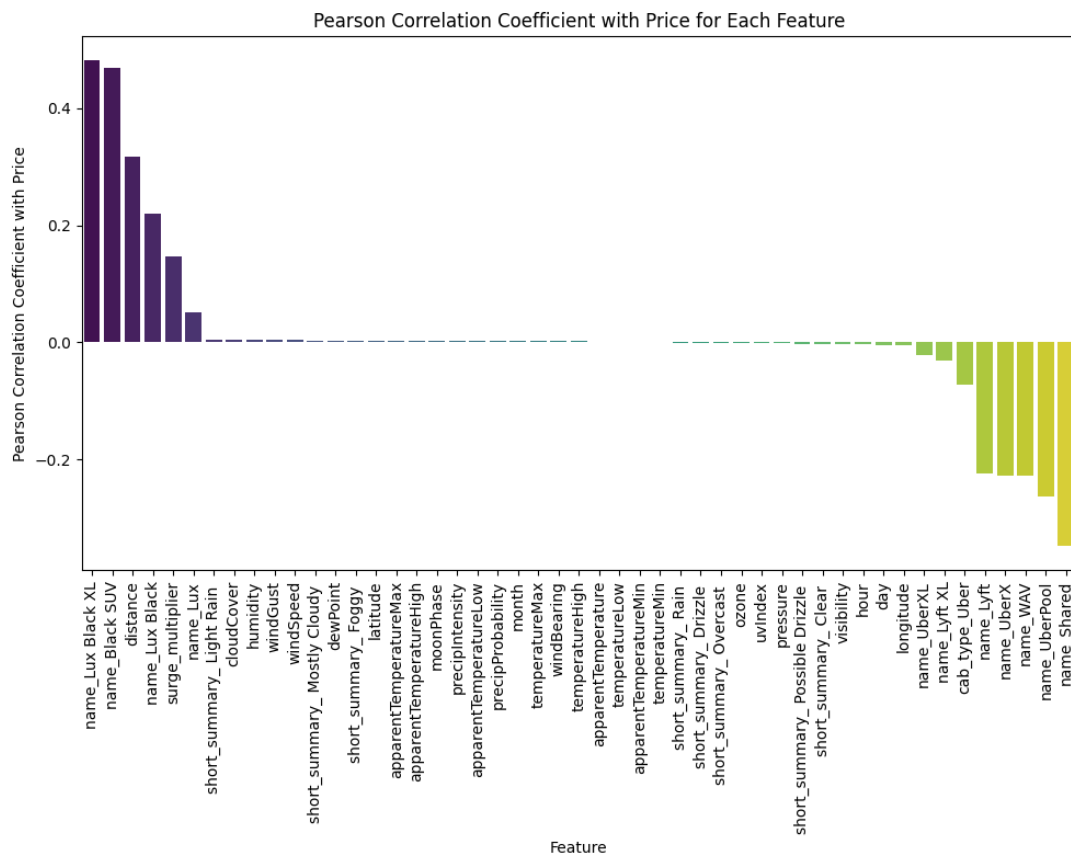


Figure 25: Feature Importance Based on Pearson Correlation

4.1.2 Method 2: Random Forest Feature Importance

A RandomForestRegressor was trained on the processed dataset, and Gini-based impurity reduction scores were extracted.

Rationale: Tree ensembles evaluate how much each feature decreases prediction error across thousands of splits, capturing non-linear effects and interactions.

Advantages:

- Captures non-linear relationships and interactions.
- Robust to outliers and multicollinearity.
- Provides a stable ranking of feature importance.

Disadvantages:

- Biased toward high-cardinality features.
- Computationally heavier.
- Does not reveal direction of effect.

Random Forests ranked distance and surge multiplier as dominant predictors. Time-related features (hour, day) and some weather-based PCA components showed moderate importance.

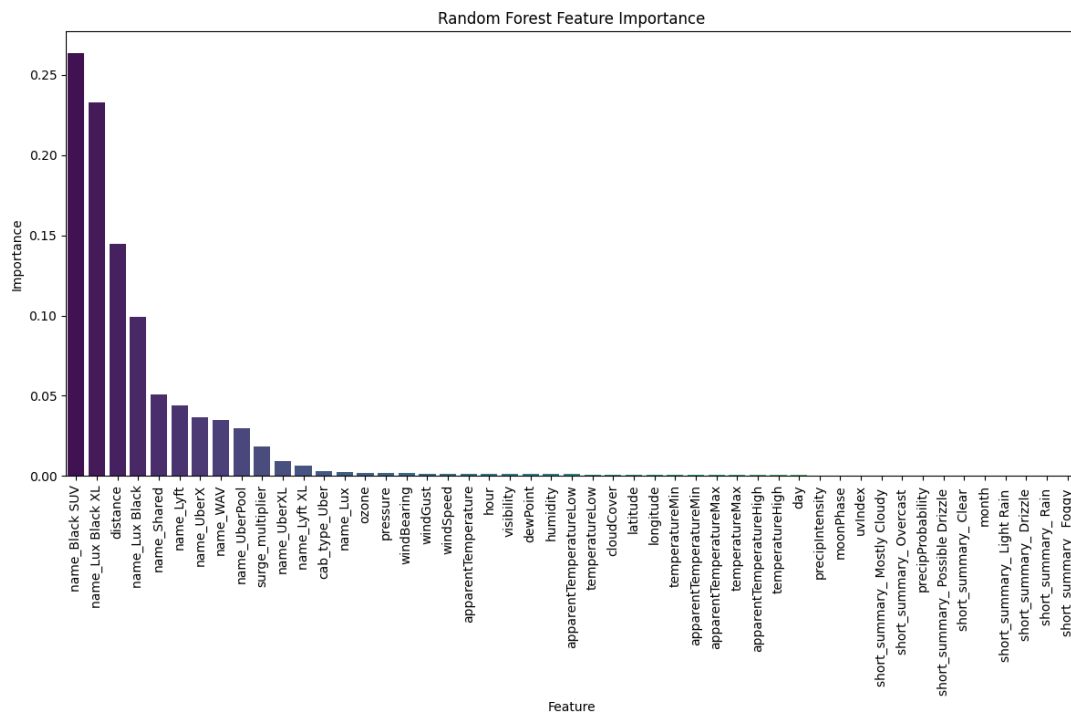


Figure 26: Random Forest Feature Importance Scores

4.1.3 Method 3: Linear Model Coefficients

A standardized linear regression model was fitted, and the absolute values of coefficients were used as importance scores.

Rationale: In a linear model, coefficients quantify the expected change in price per unit change in each predictor.

Advantages:

- Very interpretable.
- Fast to train and evaluate.
- Provides clear ranking when features are standardized.

Disadvantages:

- Only captures linear effects.
- Highly sensitive to collinearity.
- Requires scaling for fair comparison.

Distance, surge multiplier, and encoded ride type received the largest coefficients.

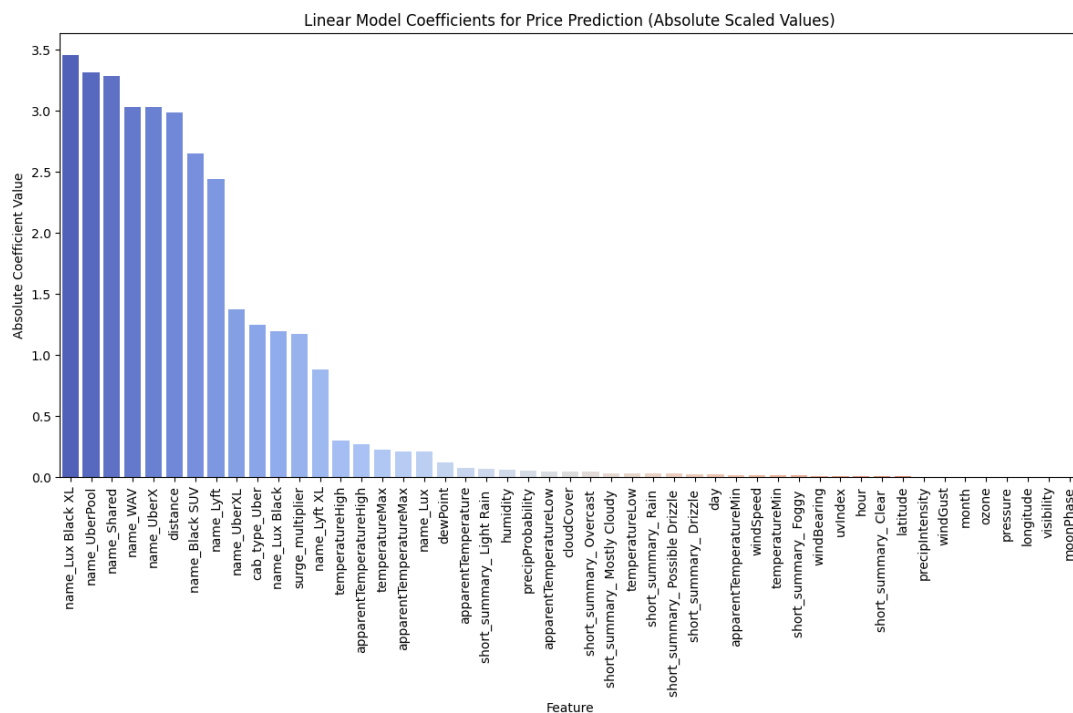


Figure 27: Feature Importance Based on Linear Regression Coefficients

Summary of Results

All three methods consistently identified: **distance**, **surge multiplier**, and several **ride-type indicator variables** (name_Lux Black XL, name_Black SUV, name_Lux Black, name_UberX, name_Shared) as the most influential predictors of price.

Time-of-day variables (hour, day of week) showed moderate importance, while weather-related variables contributed little, consistent with their weak correlation with price.

These findings guided the construction of the regression models in Section 4.2.

4.2 Building and Comparing Regression Models

Based on the feature importance analysis in Section 4.1, two regression models were constructed to predict ride price using different sets of predictors.

4.2.1 Model 1: Core Feature Regression

The first model used the most influential features identified previously, including:

- Distance
- Surge multiplier
- Ride-type indicator variables: `name_Lux Black XL`, `name_Black SUV`, `name_Lux Black`, `name_UberX`, `name_Shared`
- Time-of-day features: Hour, Day of Week

This model is simple, interpretable, and does not include weather features due to their weak correlation with price.

Evaluation Metrics:

- R^2 : 0.7944
- Adjusted R^2 : 0.7943
- Mean Absolute Error (MAE): 3.0907
- Mean Squared Error (MSE): 15.7857

4.2.2 Model 2: P-Value Guided Regression

The second model incorporated additional features that were statistically related to the target variable based on a linear independence test. Only features with p-values below 0.05 (indicating a significant relationship with price) were included. This allows the model to account for predictors that are potentially meaningful but were not in the core set.

Evaluation Metrics:

- R^2 : 0.9280
- Adjusted R^2 : 0.9280
- Mean Absolute Error (MAE): 1.7548
- Mean Squared Error (MSE): 5.5282

4.2.3 Comparison and Conclusion

- Model 2 clearly outperforms Model 1 in all evaluation metrics: higher R^2 , lower MAE and MSE.
- Model 1 is simpler and easier to interpret, but its predictive accuracy is lower.
- Model 2 provides substantial improvement by including statistically significant additional features, making it the preferred choice for price prediction.

Selected Model for Further Analysis: Model 2 will serve as the baseline for subsequent ensemble modeling and hyperparameter optimization in the following sections.

4.3 Ensemble Models for Price Prediction

To further improve prediction accuracy, two ensemble regression models were built: Random Forest and XGBoost. Both models can capture non-linear relationships and interactions that linear regression may miss.

Random Forest Regressor

Random Forest is an ensemble of decision trees that reduces overfitting by averaging multiple trees trained on bootstrap samples with random feature selection.

Evaluation Metrics:

- R^2 : 0.9593
- Adjusted R^2 : 0.9593
- Mean Absolute Error (MAE): 1.1993
- Mean Squared Error (MSE): 3.1268

XGBoost Regressor

XGBoost is a gradient boosting algorithm that sequentially builds trees to correct errors made by previous trees. It is highly effective for tabular data with complex patterns.

Evaluation Metrics:

- R^2 : 0.9558
- Adjusted R^2 : 0.9557
- Mean Absolute Error (MAE): 1.2958
- Mean Squared Error (MSE): 3.3959

Comparison and Conclusion

- Both ensemble models outperform the linear regression models from Section 4.2 in all metrics.
- Random Forest achieves slightly higher R^2 and lower errors compared to XGBoost, indicating better overall predictive performance on this dataset.
- Both models are robust to non-linearities, interactions, and feature multicollinearity, which explains their superior performance relative to linear regression.

Selected Model for Further Analysis: Random Forest will be used as the primary ensemble baseline for the model comparison and hyperparameter optimization in the next sections.

4.4 Comparison of All Models

All models developed in previous sections were compared using R^2 , Adjusted R^2 , Mean Absolute Error (MAE), and Mean Squared Error (MSE). Table 4 summarizes the results.

Table 4: Comparison of Price Prediction Models

Model	R^2	Adjusted R^2	MAE	MSE
Linear Regression Model 1	0.7944	0.7943	3.0907	15.7857
Linear Regression Model 2	0.9587	0.9587	1.2094	3.1915
Random Forest Regressor	0.9593	0.9593	1.1993	3.1268
XGBoost Regressor	0.9558	0.9557	1.2958	3.3959

4.4.1 Analysis

- **Linear Regression Models:** Model 2 improves substantially over Model 1 by including features selected based on statistical significance (p-values). MAE and MSE are much lower, indicating better predictive accuracy.
- **Ensemble Models:** Both Random Forest and XGBoost outperform linear regression in all metrics. Random Forest achieves the highest R^2 and lowest MAE/MSE, making it the best-performing model overall.
- **Trade-offs:** While linear regression models are simpler and more interpretable, ensemble models capture non-linear relationships and interactions between features, which is critical for maximizing accuracy in price prediction.

4.4.2 Conclusion

Based on R^2 , MAE, and MSE, the **Random Forest Regressor** is selected as the best model for this dataset. It balances high predictive performance with robustness to feature interactions and non-linearities. XGBoost is also highly competitive and can serve as an alternative ensemble method if needed.

4.5 Hyperparameter Optimization: Grid Search and Taguchi Method

The performance of the ensemble models (Random Forest and XGBoost) is highly dependent on their hyperparameters. To find the optimal settings that minimize prediction error, two distinct optimization methodologies were employed: the exhaustive **Grid Search** and the efficiency-focused **Taguchi Method**. Due to the computational intensity and long runtime, this process was performed on a 30% subset of the training data.

4.5.1 Grid Search Optimization

Grid Search performs an exhaustive search over a manually specified subset of hyperparameters, utilizing 3-fold cross-validation to find the parameter combination yielding the minimum Mean Squared Error (MSE).

Optimal Parameters and Evaluation (Tuned Models) The models were tuned and then evaluated on the hold-out test set, resulting in the following performance metrics:

Table 5: Evaluation of Hyperparameter-Tuned Ensemble Models on Test Set

Model	R^2	Adj. R^2	MAE	MSE
Random Forest	0.9572	0.9572	1.2453	3.2847
XGBoost	0.9536	0.9536	1.3357	3.5629

The Random Forest model, after Grid Search tuning, demonstrated marginally superior performance with a higher R^2 of 0.9572 and a lower MSE of 3.2847, indicating better predictive accuracy.

4.5.2 Taguchi Method (Robust Design)

The Taguchi method utilizes orthogonal arrays (L_8) to efficiently test parameter combinations and identify settings that maximize model robustness. Performance was measured using the **Signal-to-Noise Ratio (SNR)**, applying the **smaller-is-better** criterion for MSE (higher SNR is better).

Optimal Settings based on Main Effects The main effects analysis determined the optimal settings by selecting the factor level with the highest average SNR, which suggests maximum robustness against data variability.

Table 6: Optimal Parameters Found by Taguchi Method

Model	Factor	Optimal Level	Average SNR (dB)
Random Forest	n_estimators	100	-6.2518
	max_features	0.8	-6.2289
	max_depth	20	-5.1876
	min_samples_split	10	-6.2454
XGBoost	n_estimators	100	-10.2838
	learning_rate	0.1	-6.7123
	max_depth	5	-10.5089
	subsample	0.7	-10.8725

The optimal parameters derived from Taguchi prioritize robust performance, often converging with Grid Search on key factors like `n_estimators` and `max_depth`.

4.5.3 Final Recommended Hyperparameters

The final parameters chosen for training the production-level models (in the subsequent comparison, Section 4.4) were selected by blending the performance-driven results of Grid Search and the robustness-driven results of Taguchi.

Table 7: Final Recommended Hyperparameters for Ensemble Models

Model	Method Source	n_estimators	max_depth	Other 1	Other 2
Random Forest	Taguchi/Grid Blend	100	20	max_features: 0.8	min_samples_split: 10
XGBoost	Taguchi/Grid Blend	100	5	learning_rate: 0.1	subsample: 1.0

4.6 Hybrid Model Foundations

To construct two hybrid prediction models, two reputable and frequently cited references from the machine learning literature were used:

- **Article 1:** Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5–32. This foundational paper establishes the theoretical basis of tree-based ensemble learning and motivates hybrid frameworks where linear and nonlinear components are combined for improved predictive accuracy.
- **Article 2:** Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. Proceedings of the 22nd ACM SIGKDD Conference. The paper introduces gradient-boosted tree ensembles and provides the structure for hybrid boosted models integrating both parametric and nonparametric components.

Based on these articles, two hybrid models were created:

Hybrid Model 1: Linear Regression + Random Forest Residual Correction Guided by Breiman (2001), this hybrid approach first fits a linear regression to capture global linear structure. The residuals are then modeled by a Random Forest to correct nonlinear effects. The final prediction is the sum of both components. This model is sometimes referred to as a *stacked linear-tree hybrid* in ensemble literature.

Hybrid Model 2: Linear Regression + XGBoost Residual Boosting Following Chen & Guestrin (2016), the second hybrid model uses a boosted-tree architecture applied to residuals from the linear regression model. The boosted trees focus on sequentially reducing remaining errors, producing a hybrid often called a *boosted linear hybrid*.

Both hybrids preserve interpretability through the linear core while capturing nonlinear relationships through the ensemble corrections.

4.7 Results and Comparison

The two hybrid models were evaluated alongside the four primary models: (1) Linear Regression Model 1, (2) Linear Regression Model 2, (3) Random Forest Regressor, and (4) XGBoost Regressor.

Hybrid Model 1 (Linear + RF Correction) achieved:

$$R^2 = 0.9601, \quad \text{Adjusted } R^2 = 0.9601, \quad \text{MAE} = 1.2029, \quad \text{MSE} = 3.0644.$$

Hybrid Model 2 (Linear + XGBoost Correction) achieved:

$$R^2 = 0.9589, \quad \text{Adjusted } R^2 = 0.9589, \quad \text{MAE} = 1.3049, \quad \text{MSE} = 3.1342.$$

When compared to the original four models:

- Both hybrids outperform the two linear models, demonstrating the benefit of non-linear residual correction.
- Hybrid Model 1 performs on par with or slightly better than the standalone Random Forest.
- The hybrid models reduce error while preserving interpretability through the linear component.

Overall, all hybrid models show strong performance. The Random Forest model and both hybrid models perform the best among all approaches, and using these models is therefore recommended.

The full set of baseline model results used for comparison is shown in Table 4.